LESSON 6.1.2

Important Roles in AI

A look at the key roles that collaborate to build, deploy, govern, and secure modern AI systems from raw data to reliable production service.

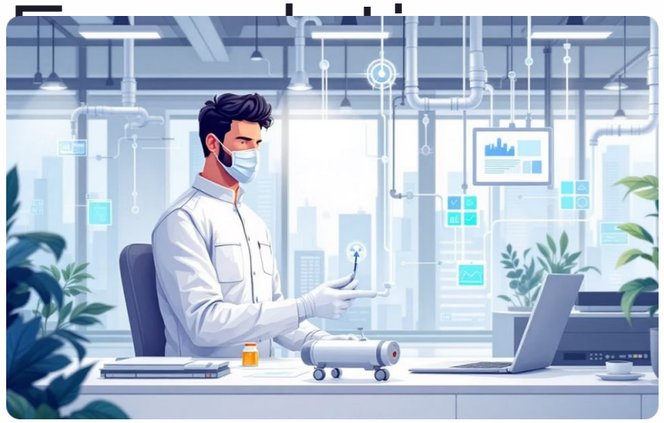
The AI Team Landscape

Modern AI programs require a diverse set of specialists working in concert, each owning a critical layer of the pipeline.

- Data & Modeling
- Engineering & Infrastructure
- Security & Governance
- Risk & Audit

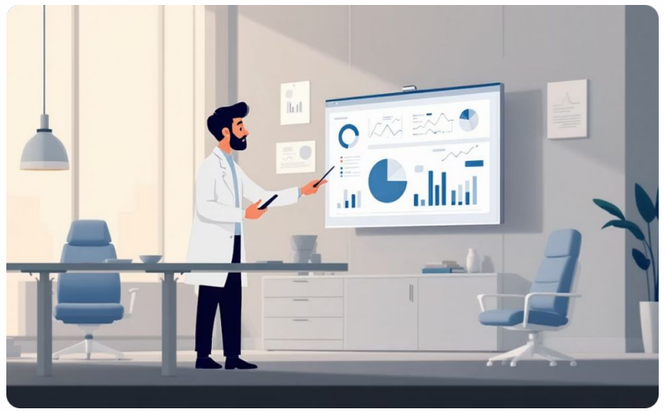
DATA & MODELING

Building the



Data Engineer

Builds and maintains pipelines that **acquire, clean, label, and deliver** high-quality data with appropriate controls.



Data Scientist

Explores data to **define the problem**, select features, and develop baseline models that demonstrate value.

ENGINEERING

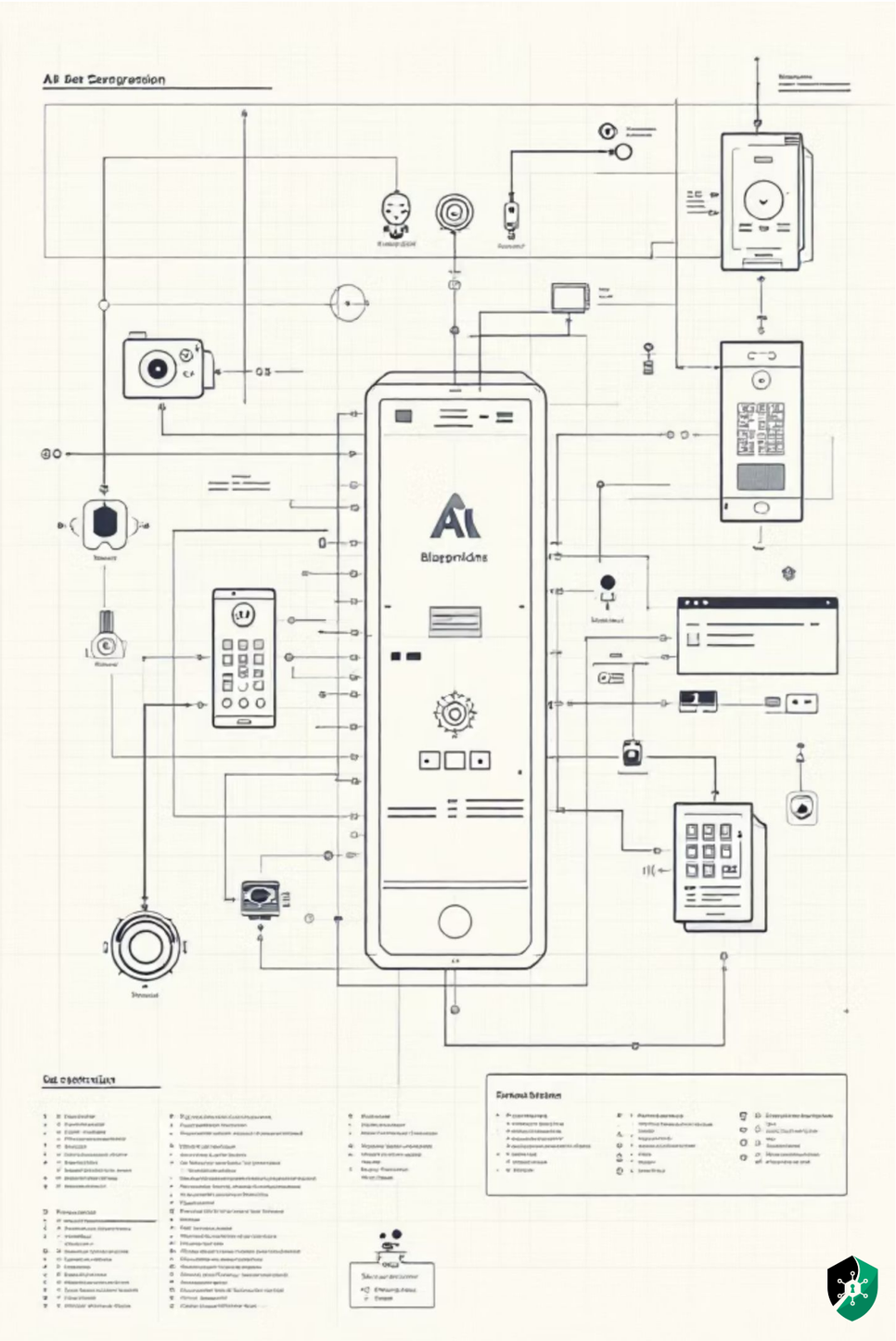
From Prototype to Production

ML Engineer

Transforms prototypes into **production-grade models**, optimizing code, packaging dependencies, writing inference services, and designing evaluation tests.

AI Architect

Designs how all components fit within the enterprise, choosing patterns for **data flows, model serving, and system integration** to ensure scalability and resilience.



INFRASTRUCTURE

The Platform & MLOps Layer

Platform Engineer

Provides shared infrastructure: compute clusters, storage, feature stores, model registries, and observability tools for safe experimentation and deployment.

MLOps Engineer

Automates the model lifecycle — CI/CD for models and prompts, versioning, lineage, retraining workflows, drift monitoring, and rollback triggers.

SECURITY

Protecting AI Systems

The **AI Security Architect** designs defenses against threats including:

🔒 Data Leakage

☠️ Model Poisoning

💉 Prompt Injection

🚫 Abuse & Misuse

Safeguards include secrets management, access control, isolation, content filtering, and adversarial testing.

GOVERNANCE & RISK

Policy, Risk & Accountability

AI Governance Engineer

Codifies approval workflows, manages deployment risks, collects required evidence, and enforces retention and audit requirements.

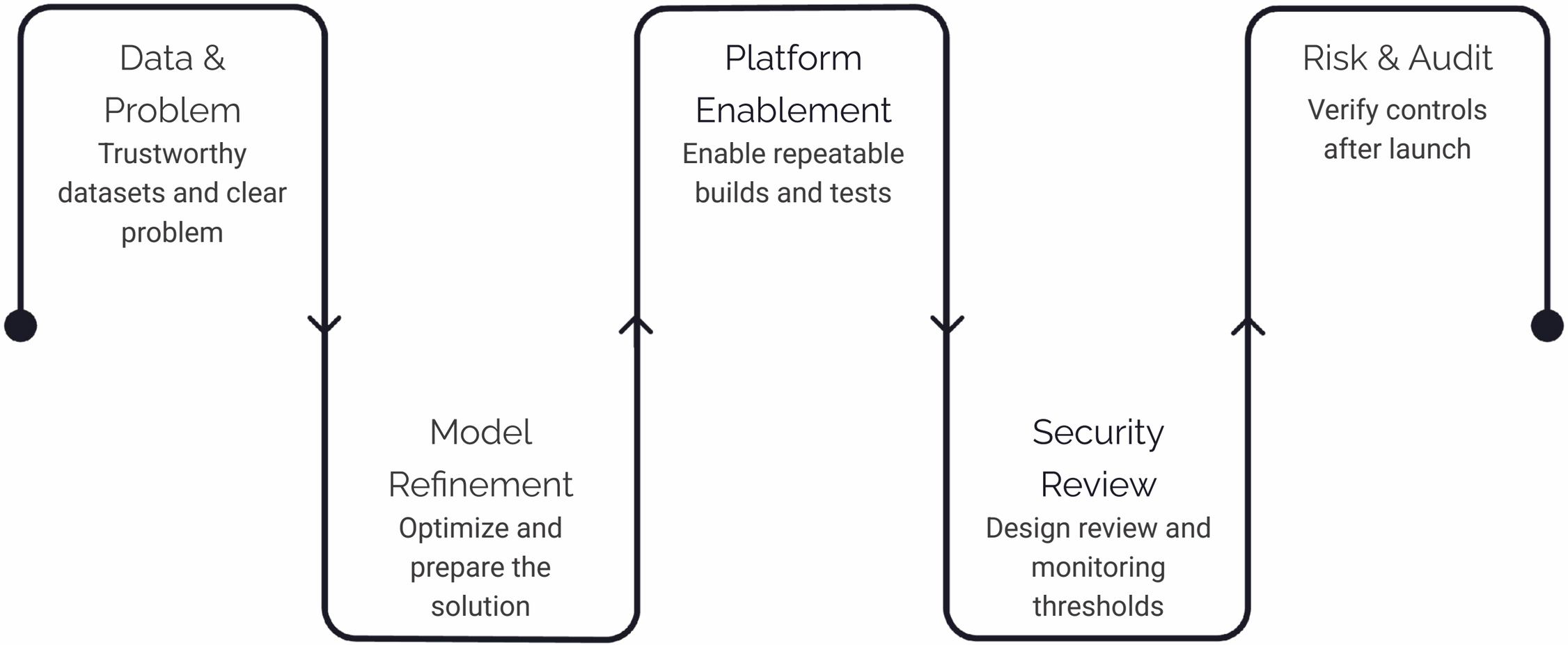
AI Risk Analyst

Assesses use cases on legal, ethical, and business criteria. Recommends mitigations and tracks key indicators like override rates and error impacts.

AI Auditor

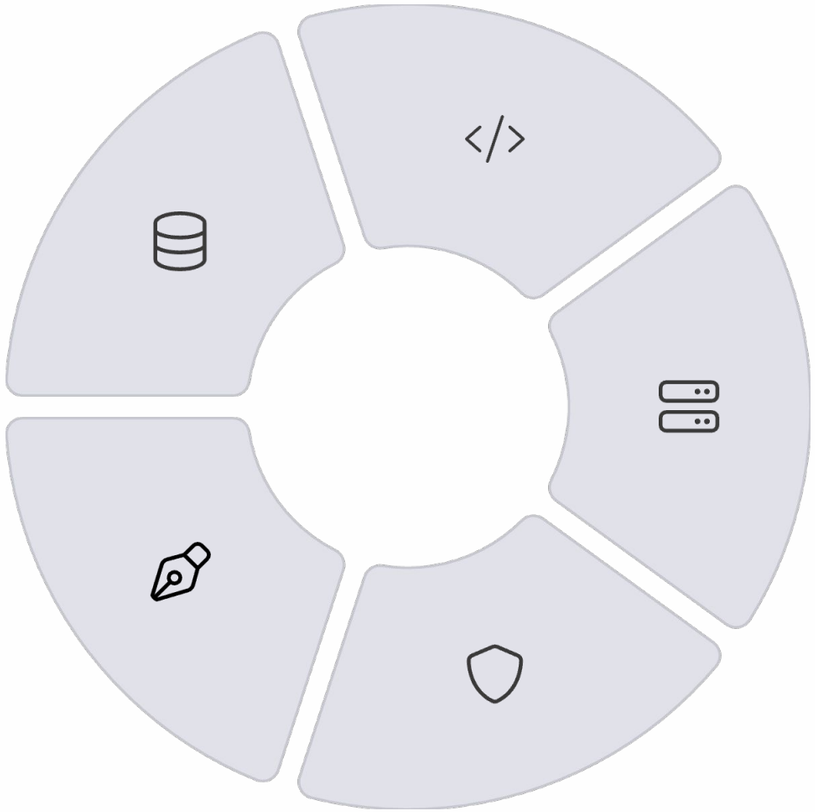
Provides independent assurance, sampling artifacts, recreating decisions, verifying approvals, and confirming monitoring operates as documented.

The Handoff: Idea to Live Service



Each role passes a verified artifact to the next, ensuring full traceability from raw data to a reliable, governed production service.

Roles Work as One System



Data Layer

Engineer + Scientist

Model Layer

ML Engineer + Architect

Platform Layer

Platform + MLOps

Security Layer

AI Security Architect

Governance Layer

Governance + Risk + Audit

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Key Takeaways

1 **Many roles, one mission**

Data engineers, scientists, ML engineers, architects, platform and MLOps engineers each own a distinct, critical layer.

2 **Security is built-in, not bolted on**

The AI security architect protects against leakage, poisoning, prompt injection, and abuse throughout the pipeline.

3 **Governance enforces accountability**

Governance engineers, risk analysts, and auditors ensure policy, evidence, and monitoring operate as required.

4 **Handoffs ensure traceability**

Each stage passes verified artifacts to the next, creating a fully traceable path from raw data to live, governed service.